

How modern computers compute

How can an object made from millions of tiny, complex parts produce outputs like words, music, art, or motion?

Displaying data

A computer is a machine for manipulating numbers, and to a computer, everything is numbers. Letters, symbols, sounds, and images are all represented by binary numbers. To most humans however, binary is just a string of meaningless 1s and 0s. How have computer scientists enabled computers to display the data they work with in ways that people can understand?

Image

Computer screens are made up of tiny areas called pixels that are lit up to show shapes and colours. To display a pixel as white, it's lit with equal amounts of red, green, and blue light. Other colours are obtained by mixing different proportions of these three, with black being an absence of light.

Each colour is made of three binary values for the amount of red, green, and blue in it.

00000000
00000000
11111111

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IN DEPTH

Hexadecimal

Most people find binary numbers difficult to work with. The hexadecimal system is based on multiples of 16 and uses the digits 0 to 9 followed by the letters A to F. A 24-bit binary number defining a colour can be written as six hexadecimal digits, making life easier for programmers.

FFC0FF	8AE1FF	00F1DD	00F396	C9E151
FE9AFF	4DCAFF	00D4C3	00DA86	B2C848
F68DFF	2CC0FF	00CAB9	00CF80	A9BD44

Text

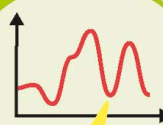
Letters and characters are represented by a standardized set of binary numbers. The operating system on a computer contains code that can translate these binary numbers into a pattern of pixels on a screen. Word processors and web browsers contain code for the pixel patterns of various fonts.

S

01010011

Sound

Computers produce sounds by translating binary numbers into electrical signals. These electrical signals are then fed into a loudspeaker that makes a very thin sheet of metal vibrate. The vibrations create pressure changes in the air that human ears interpret as sound.



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